Integrated Circuit Biasing

Moving away from discrete circuit intuition (where resistors are cheap and transistors are expensive)

to

analog IC design reality (where transistors are virtually free, but resistors consume valuable silicon area).

Discrete Design vs. **IC Design**

- **Discrete Design:** Use big resistors and capacitors to set operating points
- **IC Design:** A 100 kΩ resistor takes up more chip area than 50 transistors!
- **Core Objective:** We need circuit topologies that generate stable, precise bias currents using **mostly transistors** and as few small resistors as possible.

Integrated Circuit Biasing

- A current source is not a “naturally” occurring device. It can be simulated by a network of transistors and circuit elements.
- Integrated-circuit fabrication technology imposes constraints
  - large capacitors are not available
  - very small capacitors are easy to fabricate
- One objective is to realize as many functions as possible using transistors only.
- Reduction of device size is of great concern.

Discrete and Integrated Circuits

A discrete circuit is constructed of components that are manufactured separately. Later, these components are connected together, by conductors like wires, in a circuit board or a printed circuit board (PCB).

On the other hand, in an integrated circuit, the components and their inter-connections are manufactured concurrently by a sequence of processing steps.

The types of components that are available and their practical values depend heavily on the approach taken for implementation. (for example, the capacitors in discrete circuits can be in the range of 1pF to 1F, but only 1pF to 100pF in ICs. Also, inductors are almost impractical in ICs. But in ICs, matching of components is much easier)

Applications of discrete circuits will persist especially for some special circuits that are to be mass produced, but today the bulk of electronic systems are based on ICs.

Processing steps in manufacturing ICs incur cost and failures and are usually different for different technology. BJT technology are used more for high-quality analog circuits, while MOS are more for general analog circuits and digital circuits. Today, semiconductor industry can manufacture both BJT and MOS on the same chip, called BiCMOS technology.

Why current source needed for IC biasing?

- Resistors take too much space on the chip. Source degeneration with resistor is not implemented in ICs.
- The goal of a good bias is to ensure drain current (I_D) and Drain source voltage V_{DS} would not change (e.g., due to temperature variation). One can force I_D to be constant using a current source.

Introduction

- Integrated circuits have transistors which are manufactured simultaneously with the same device parameters (parameters from chip to chip will vary)
- As a result, different bias techniques are employed than in discrete designs
- One common technique is current source biasing, which allows the designer to take advantage of matched devices
- We will begin by looking at some simple current source circuits

IC Biasing

- Biasing in IC design is based on the use of constant- current sources.

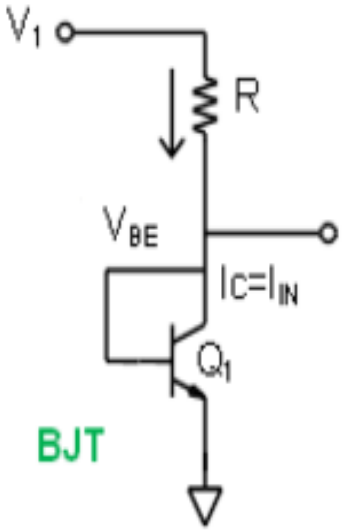
- A constant dc current (called a reference I) is generated at one location and is then replicated at various other locations for biasing the various stages through a process known as current steering.

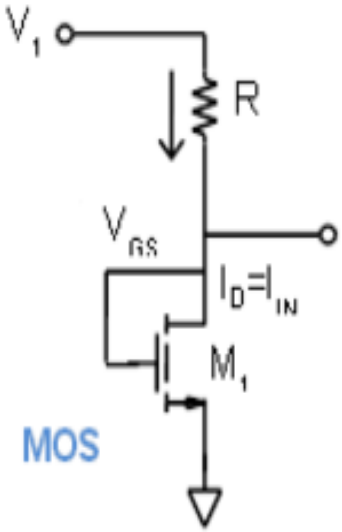
Current source

- Two types of basic gain cells exist:
  - Common-source (CS)
  - Common-emitter (CE)
- Both are loaded with constant-current source.
  - This is done because of difficulties associated with fabrication of exact resistances.
  - It also facilitates increased gain.

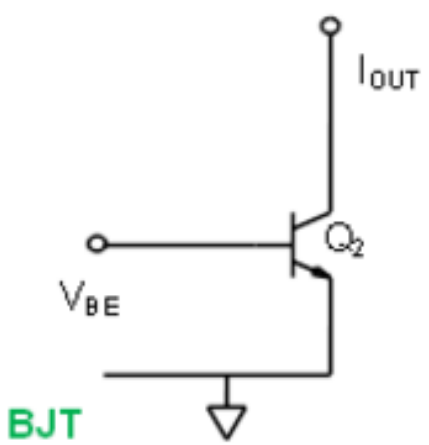
Discrete vs Integrated Circuits

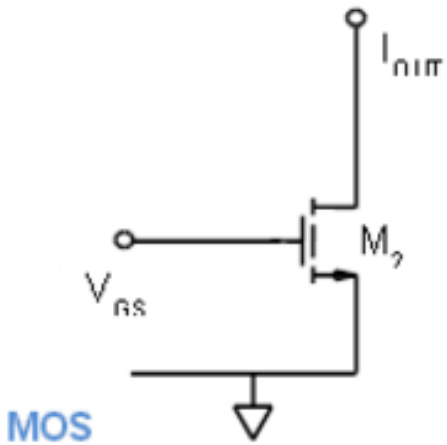
Discrete Circuits	IC
• Resistors and capacitors are frequently used	• Use mostly transistors – Resistors and capacitors occupy too much areas
• AC coupled with coupling capacitors	• Mostly DC coupled (without capacitors)
• Large DC power supply voltage	• Low DC power supply voltage (~ 1V)
• Transistor choice limited to available parts	• Can vary device size
• Mostly BJT, some MOS	• Predominantly CMOS – BiCMOS provides BJT



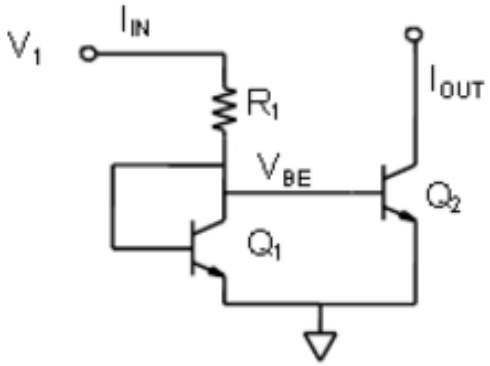


Current to voltage convertor





voltage to current convertor

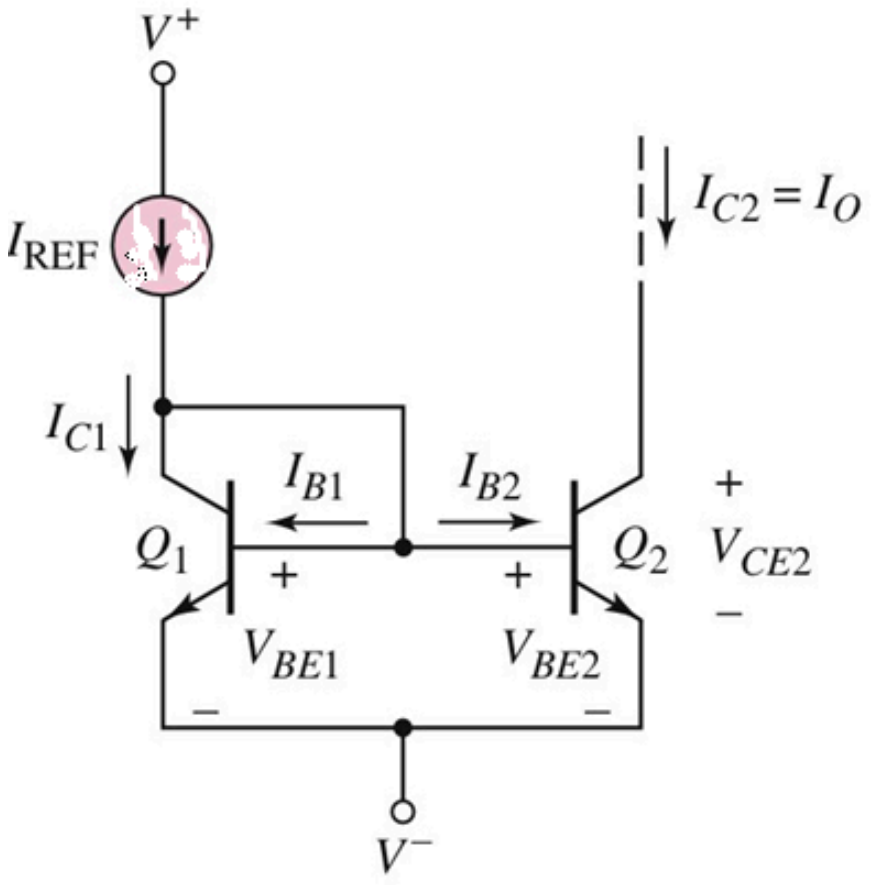


- Modern integrated circuits use MOSFETs extensively
  - Very high densities of transistors – up to 10^9 transistors/cm^2 in some ULSI memory arrays.
  - Off-chip discrete resistors and capacitors are ***NOT*** commonly used
  - On-chip resistors and capacitors generally ***small***
  - Multistage amplifiers are usually ***DC-coupled***
- Transistors used wherever possible to implement current sources, resistors, capacitors,

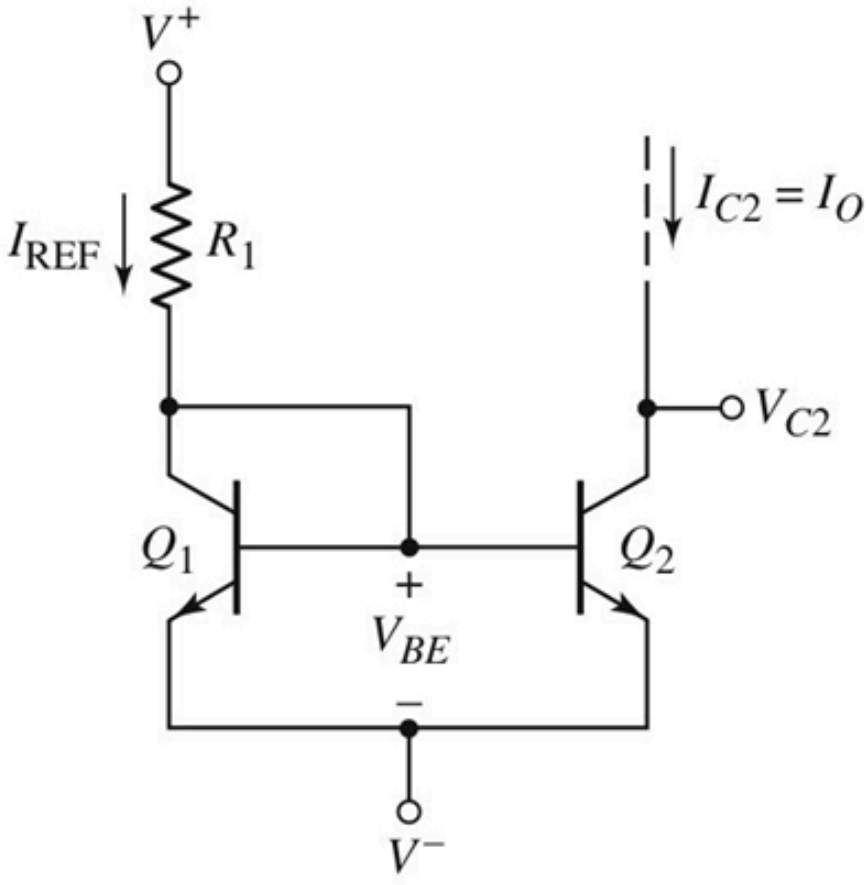
Two transistor Current source

- Also known as Current mirror
- Consists of two matched or identical transistors Q_1 and Q_2, operating at same temperature with their base terminals and emitter terminals connected together.
- The B-E voltage is same in two transistors.
- Q1 is connected as diode
- When supply voltage is applied, B-E junction of Q_1 is forward biased and reference current I_{ref} is established.

Two transistor Current source



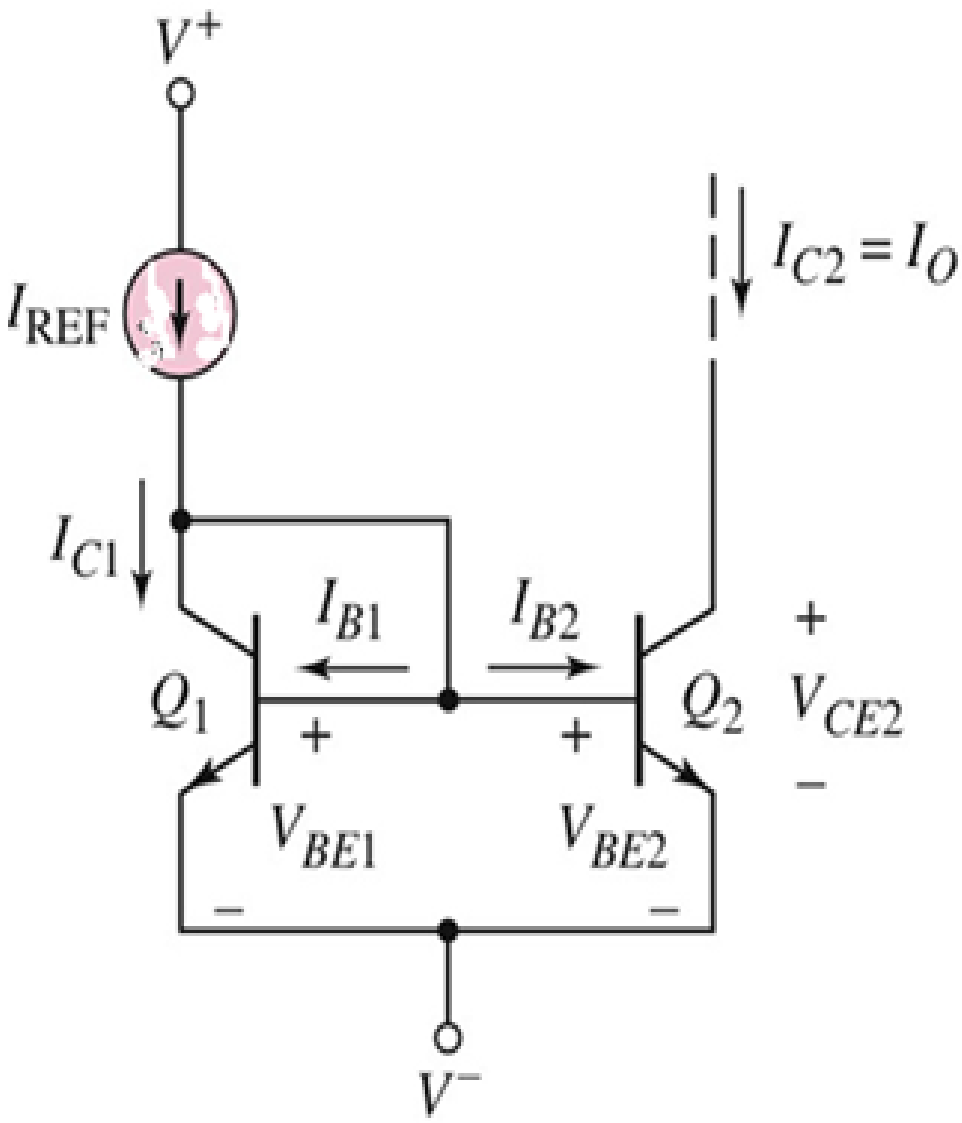
(a)



(b)

Two transistor Current source

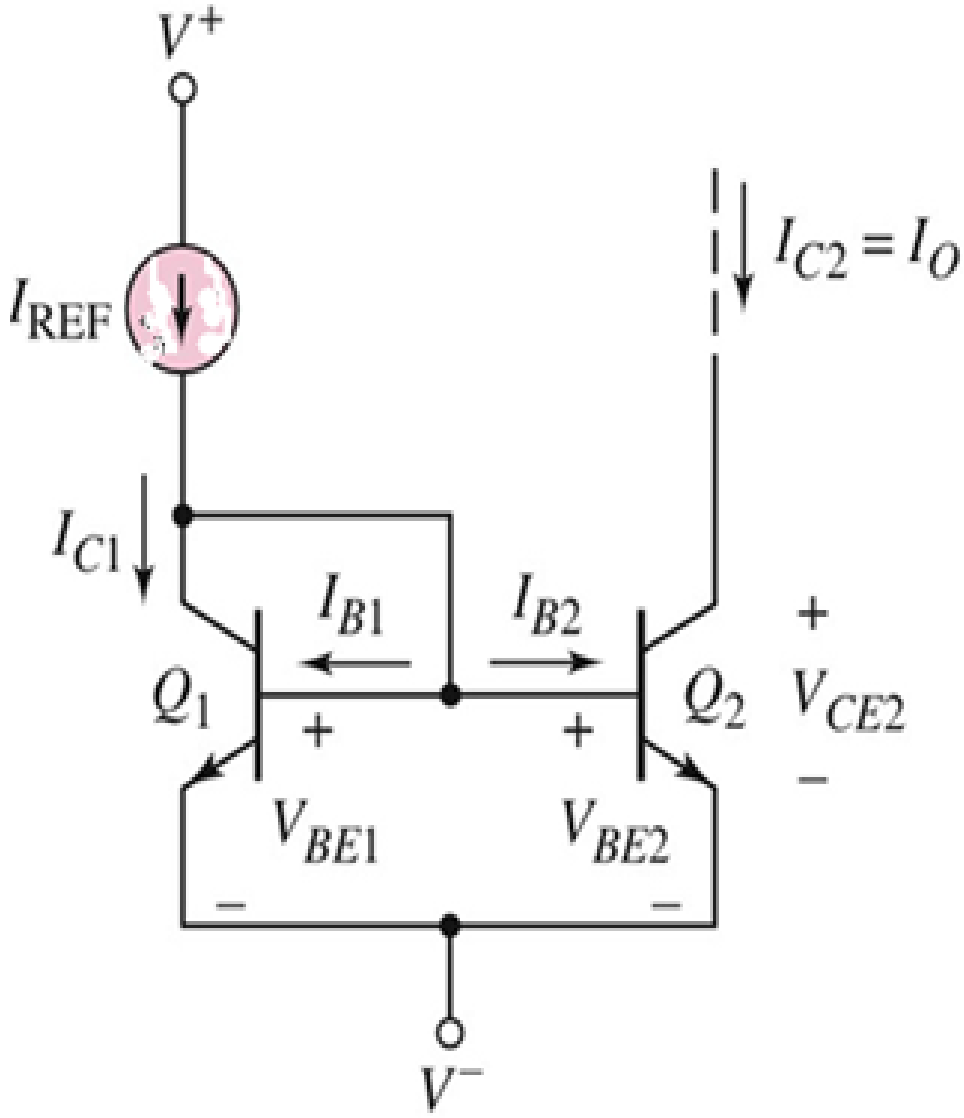
- Q_1 has its collector tied directly to its base making it a "diode-connected" transistor.
- This connection forces Q_1 into the **forward-active region** because $V_{CB1} = 0$ V, which keeps the junction out of saturation.



Two transistor Current source

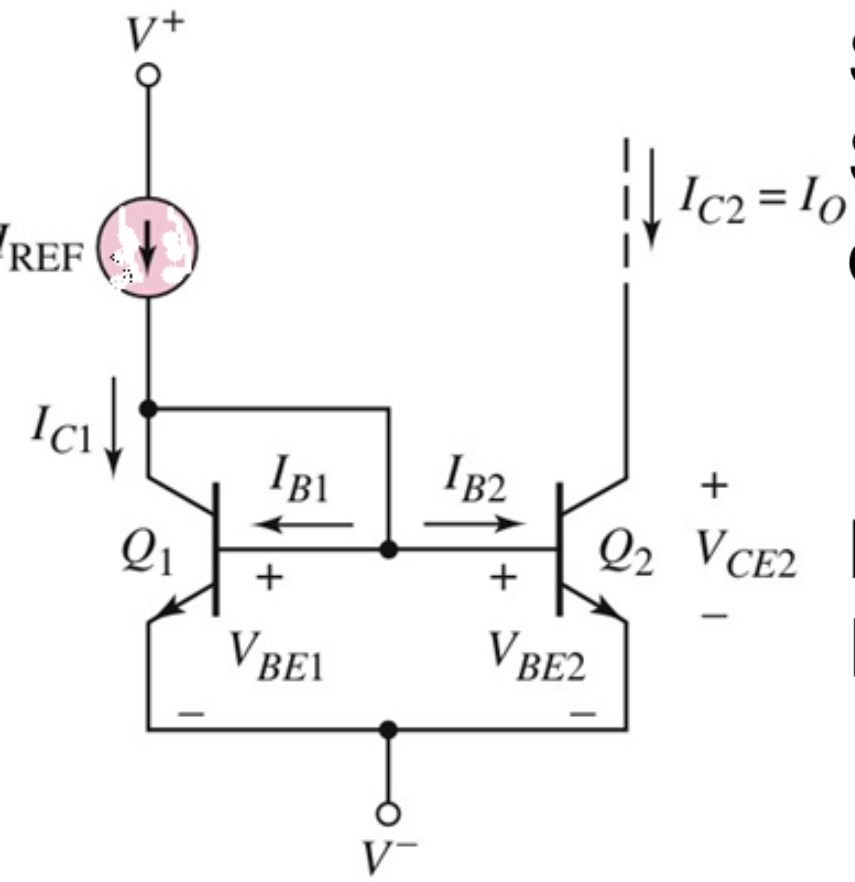
In a diode-connected BJT, $V_{CE} = V_{BE} \approx 0.7$ V

Since 0.7 V is significantly higher than the saturation threshold $V_{CEsat} \approx 0.2$ V, the transistor is locked into active mode and can never saturate.



Two transistor Current source

- Once V_{BE1} is established, it is applied to B-E junction of Q_2.
- Applied V_{BE2} turns Q_2 on and generates load current I_O. (which is used to bias transistor or transistor circuit)
- Reference current can be set by connecting resistor to the positive voltage source as shown in fig.
- $I_{REF} = (V^+ - V_{BE} - V^-)/ R_1$



(a)

V_{BE} is same in both devices.

Transistors are identical.

So, $I_{B1} = I_{B2}$: $I_{C1} = I_{C2}$

Sum of current at collector node of Q1,

$I_{REF} = I_{C1} + I_{B1} + I_{B2}$

$= I_{C1} + 2 I_{B2}$

Replace I_{C1} by I_{C2} and $I_{B2} = I_{C2}/\beta$

$I_{REF} = I_{C2} + 2 (I_{C2}/\beta)$

$= I_{C2} (1 + \frac{2}{\beta})$

$$I_{C2} = \frac{I_{REF}}{(1 + \frac{2}{\beta})}$$

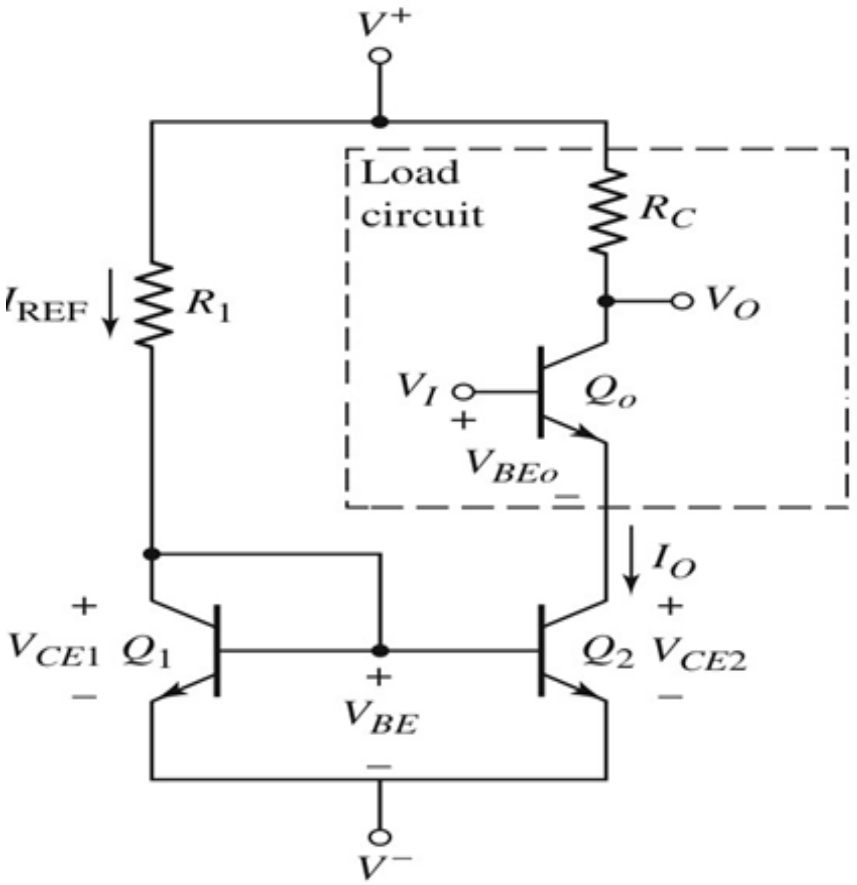
Assumption : Early voltage is infinite

DC Equivalent Circuit: 2-Transistor Current Source

- Earlier Assumption: Early voltage is infinite. So, that $r_O = \infty$
- In actual transistors Early voltage is finite which means that collector current is function of collector-emitter voltage

$$I_C = I_S e^{V_{BE}/V_T}\left(1 + \frac{V_{CE}}{V_A}\right)$$

DC Equivalent Circuit: 2-Transistor Current Source



Voltage V_I, applied at base of Q_O is DC voltage.

If V_I changes V_{CE2} changes.(as V_{BEo} is essentially constant)

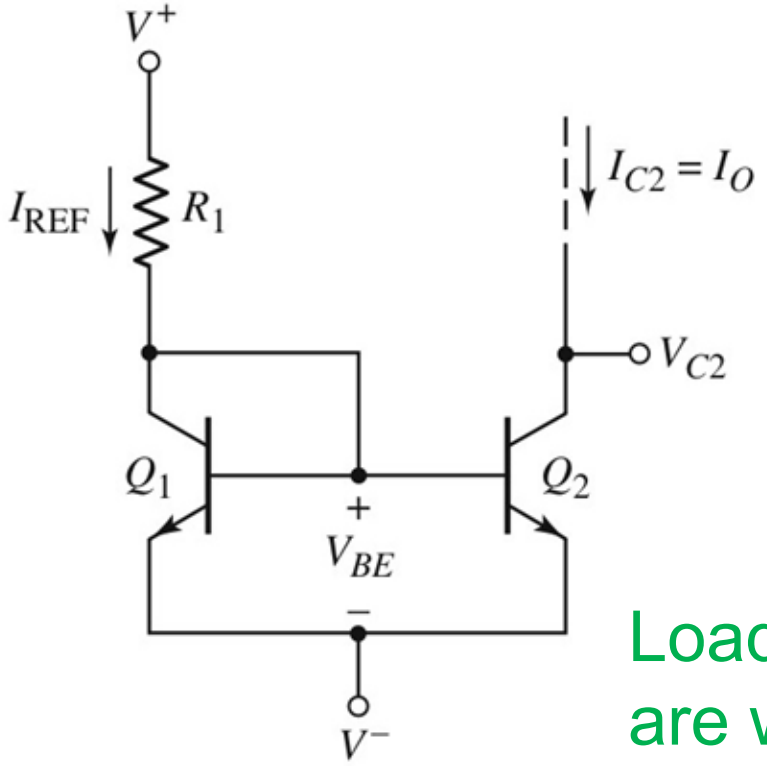
V_{CE2} Changes , I_O changes.(Early Effect)

$$\frac{I_O}{I_{REF}} = \frac{1}{1+\frac{2}{\beta}} \times \frac{1+\frac{V_{CE2}}{V_A}}{1+\frac{V_{CE1}}{V_A}}$$

$V_{CE1} = V_{BE}$ = Constant

$V_{CE2} = V_I - V_{BEo} - V^-$

If V_I changes, V_{CE2} changes

- Circuit to be designed has configuration shown in figure. Assume matched transistors are available with parameters $V_{BE(on)} = 0.6$ V, $\beta = 100$, $V_A = \infty$. Designed output I_O is to be 200 μA. Find R_1. The bias voltages are to be $V^+ = 5V$ and $V^- = 0$.
- 

- $I_{REF} = I_O (1 + (2/\beta))$
- $= 204\ \mu A$
- $R_1 = (V^+ - V_{BE} - V^-)/$
- $= 21.6\ K\Omega$

Load Current and reference current are with 2% of each other.

Finite Output resistance R_O

Load current is function of output resistance.

$$\frac{dI_0}{dV_{CE2}} = \frac{I_{REF}}{1+\frac{2}{\beta}} \times \frac{1}{V_A} \times \frac{1}{1+\frac{V_{BE}}{V_A}}$$

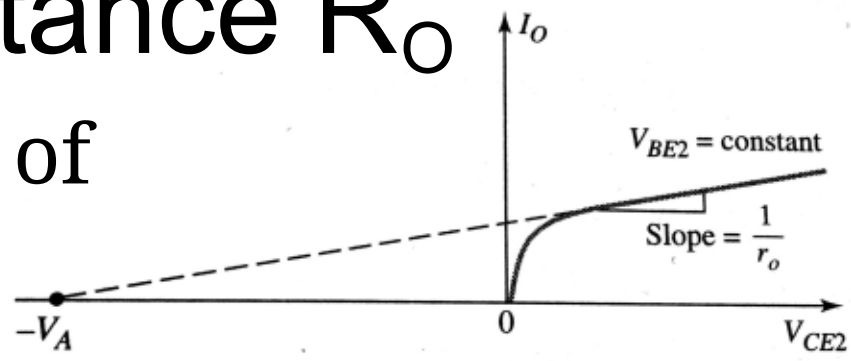


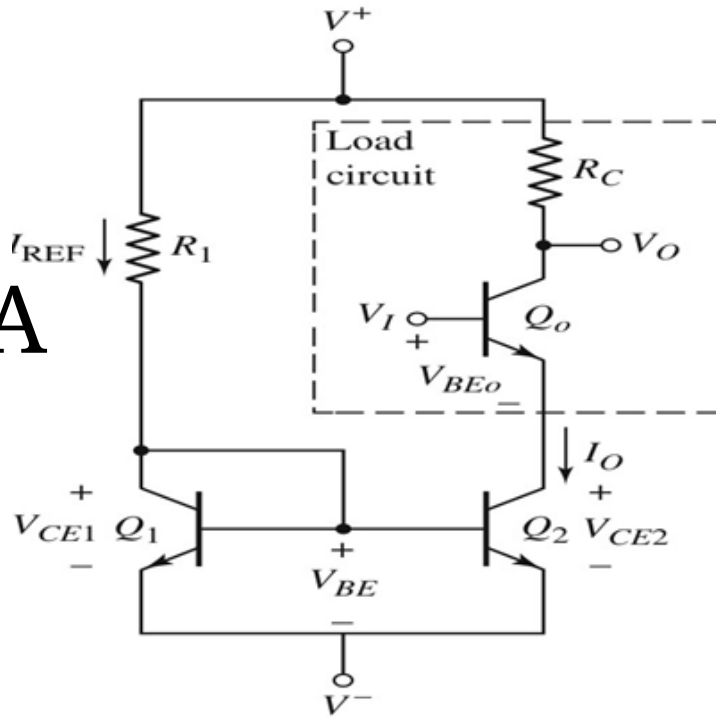
Figure 10.4 Output current versus collector–emitter voltage, showing the Early voltage

If we assume, $V_{BE} \ll V_A$, then

$$\frac{dI_0}{dV_{CE2}} \cong \frac{I_O}{V_A} = \frac{1}{r_o}$$ where r_O is small signal output resistance looking into collector of Q_2.

Example 10.2

- V^+ =5 V, V^- = -5 V R_1 = 9.3 KΩ Assume transistor parameters are β = 50, $V_{BE(ON)}$=0.7 V, V_A = 80 V. Determine change in I_O as V_{CE2} changes from 0.7 V to 5 V
- $I_{REF} = (V^+ - V_{BE} - V^-)/ R_1 = 1$ mA
- For V_{CE2} = 0.7 V,
- $I_O = I_{REF}/(1 + (2/\beta)) = 0.962$ mA
- $r_O = V_A / I_O = 83.2$ KΩ

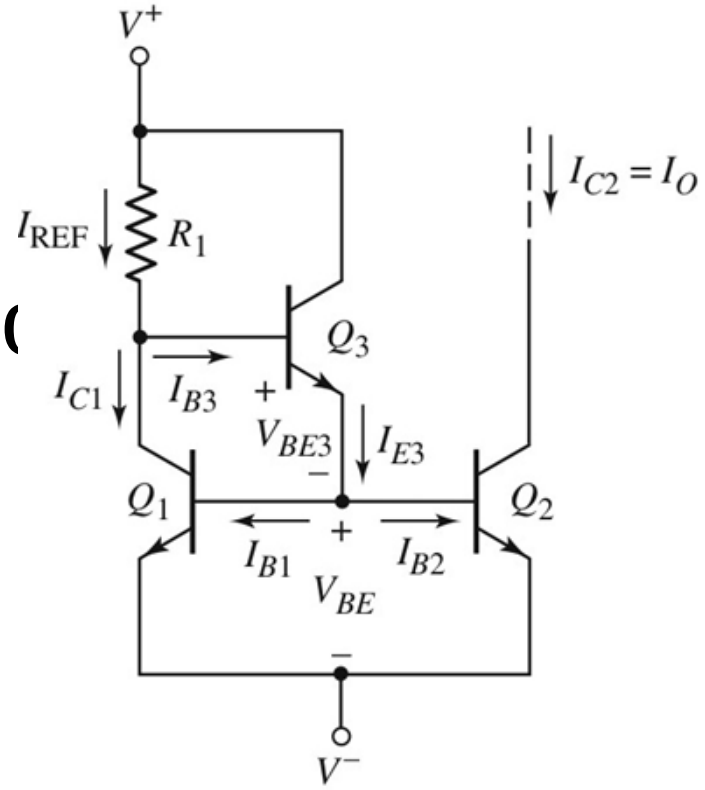


- $d I_O / dV_{CE2} = 1 / r_O$
- $d I_O = (1 / r_O) \, dV_{CE2}$
- $= 0.052 mA$
- percent change in output current is
- $d I_O / I_O = 0.052/0.962 = 0.054 = 5.4\ \%$

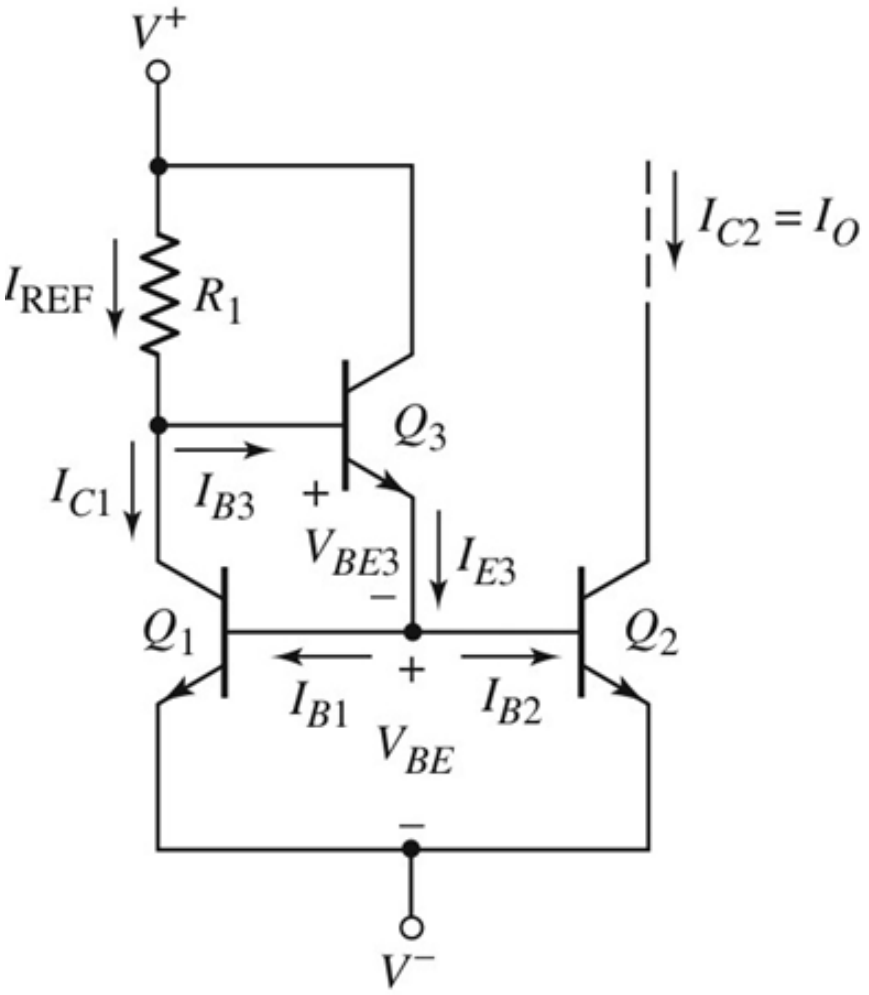
In many circuits 5% change in bias current may be insignificant, but in D-A converters bias current may be held to very tight tolerance.

Improved Current-Source Circuits

- Improved load current stability against changes in β and changes in output collector voltage.
- Three transistors current s



Three Transistor Current Source



V_{BE} is same in both devices. Transistors are identical.

So, $I_{B1} = I_{B2}$

$I_{C1} = I_{C2}$

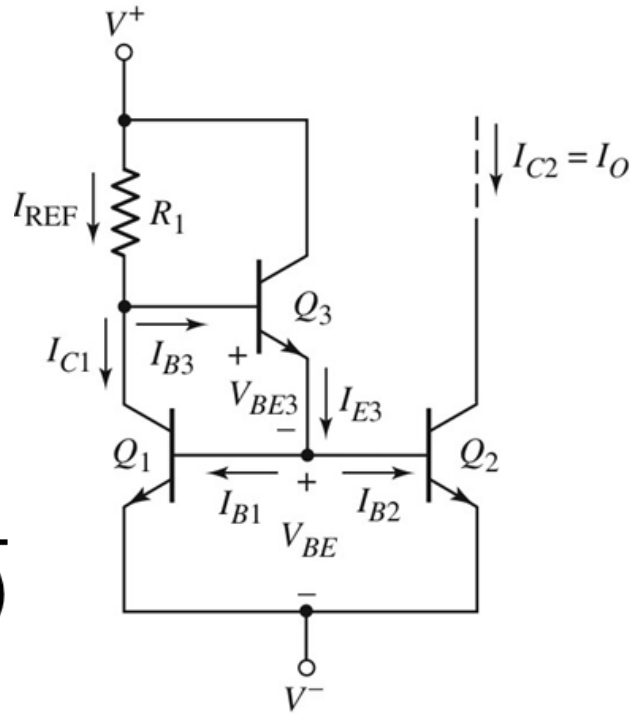
Current gain of Q_1,Q_2 as $\beta_1 = \beta_2 = \beta$

Current gain of Q_3 as β_3

Apply KCL at collector node of Q1

$I_{REF} = I_{C1} + I_{B3}$

- $I_{B1} = I_{B2}$
- $2\ I_{B2} = I_{E3}$
- $I_{E3} = (1 + \beta_3) I_{B3}$



- $I_{REF} = I_{C1} + \dfrac{I_{E3}}{(1 + \beta_3)} = I_{C1} + \dfrac{2\ I_{B2}}{(1 + \beta_3)}$
- Replace I_{C1} by I_{C2}, $I_{B2} = I_{C2} / \beta$
- $I_{REF} = I_{C2} + \dfrac{2\ IC_2}{\beta\ (1 + \beta_3)} = I_{C2} \left(1 + \dfrac{2}{\beta\ (1 + \beta_3)} \right)$
- $I_{C2} = I_O = \dfrac{I_{REF}}{\left(1 + \dfrac{2}{\beta\ (1 + \beta_3)} \right)}$
- $I_O = (V^+ - V_{BE} - V_{BE} - V^-) / R \sim (V^+ - 2V_{BE} - V^-) / R$

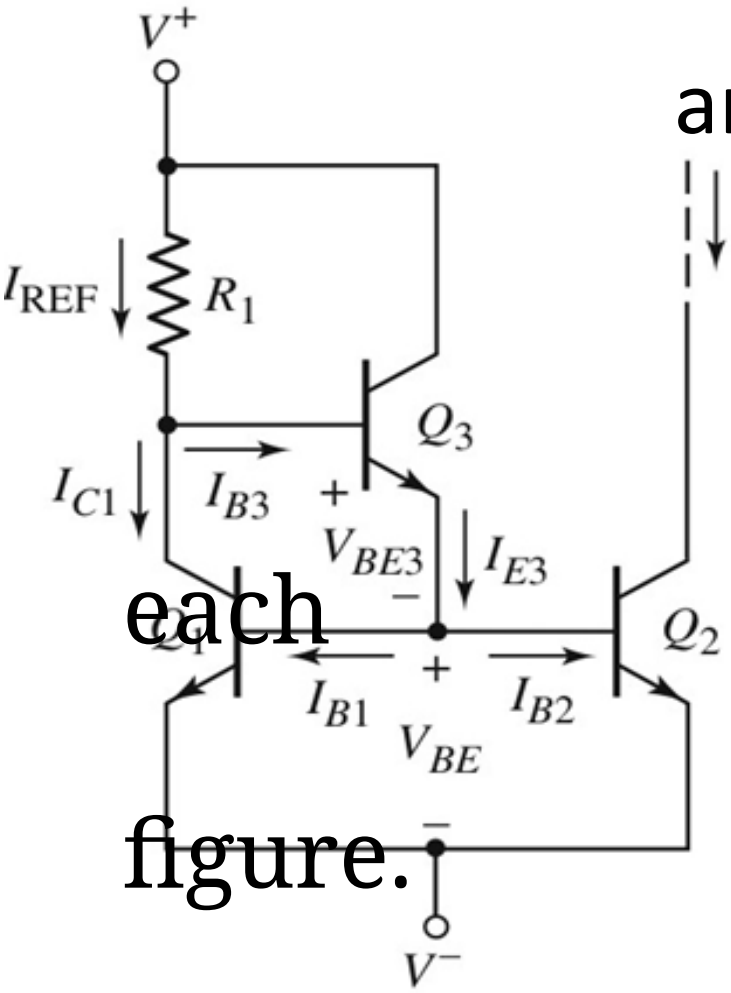
- $I_{C2} = I_O = \dfrac{I_{REF}}{\left(1 + \dfrac{2}{\beta\,(1+\beta_3)}\right)}$

------ Three transistor current source

- $I_{C2} = \dfrac{I_{REF}}{(1 + \frac{2}{\beta})}$------Two transistor current source

Exercise—Ex10.3

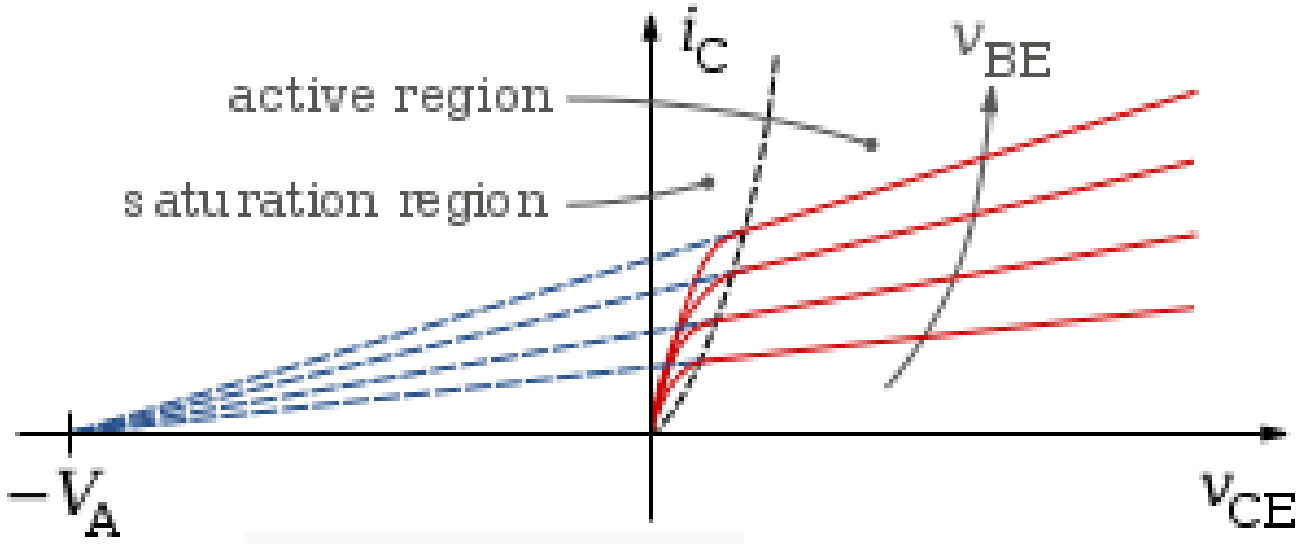
- Parameters for circuit shown are V^+ =9 V, V^- = 0



and R_1 = 12 KΩ . Assume transistor parameters are β = 75, $V_{BE(ON)}$=0.7 V, V_A = ∞. Calculate value of each current shown in figure.

- $I_{REF} = (V^+ - V_{BE3} - V_{BE} - V^-)/ R_1 \cong (V^+ - 2V_{BE} - V^-)/ R_1$
- = (9-1.4) / 12 K
- = 633.3μ A
- $I_{C2} = I_O = \frac{I_{REF}}{(1 + \frac{2}{\beta(1+\beta_3)})}$ = 6333/$(1 + \frac{2}{75*(76)})$
- = 633.1 μA
- $I_{B1} = I_{B2} =$ 8.44μA $I_{E3} = 16.88$ μA
- I_{B3}=0.225μA

Early Effect



Infinite R_O



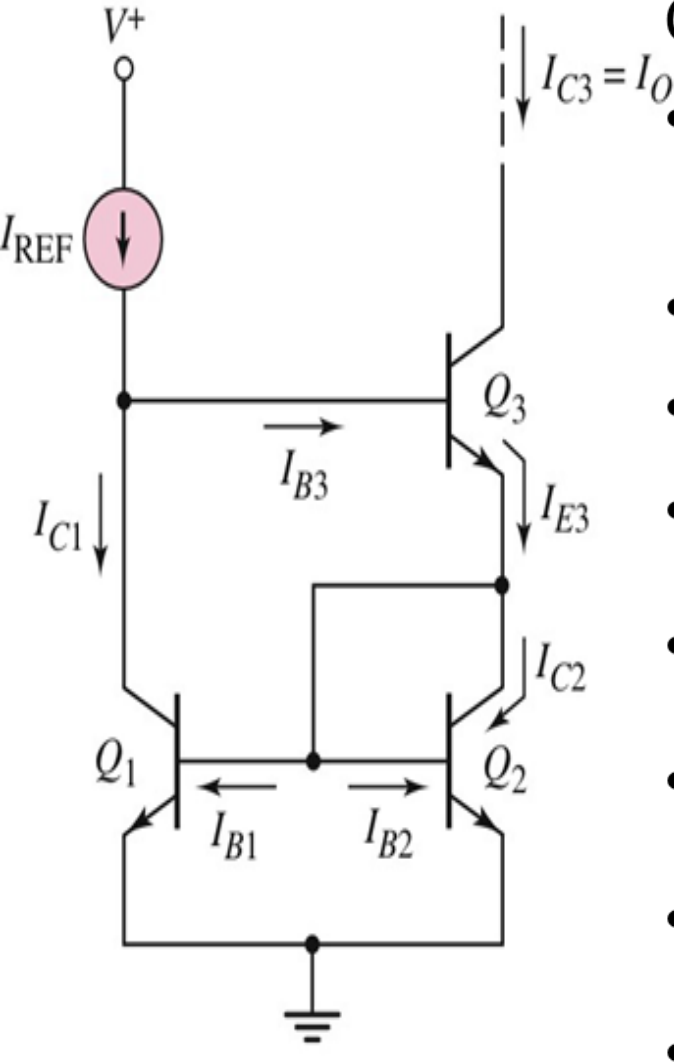


Finite R_O

$$r_o = \frac{V_A + V_{CE}}{I_C}$$

approximation $r_O = V_A / I_C$

Wilson Current source



- Transistors are identical. So, $I_{B1} = I_{B2}$ $I_{C1} = I_{C2}$

Current gain of Q_1, Q_2 as $\beta_1 = \beta_2 = \beta$

- Nodal equation at collector of Q_1, $I_{REF} = I_{C1} + I_{B3}$ ---------A
- Nodal equation at emitter of Q_3,
- $I_{E3} = I_{C2} + 2I_{B2} = I_{C2}(1+(2/\beta))$
- $I_{C2} = I_{E3} / (1+(2/\beta))$
- $= \frac{1}{1+(2/\beta)} \times \frac{1+\beta}{\beta} I_{C3} = \frac{1+\beta}{2+\beta} I_{C3}$
- Replace I_{C1} by I_{C2} in A
- $I_{REF} = I_{C2} + I_{B3} = \frac{1+\beta}{2+\beta} I_{C3} + \frac{I_{C3}}{\beta} = I_{C3}\left(\frac{1+\beta}{2+\beta} + 1/\beta\right)$
- $\mathbf{I_{C3} = I_O = I_{REF}}$

Wilson Current Source

- $I_{REF} = I_{C2} + I_{B3} = \frac{1+\beta}{2+\beta} I_{C3} + \frac{I_{C3}}{\beta}$
- $I_{C3} = I_O = I_{REF} \times \frac{1}{1+\frac{2}{\beta(2+\beta)}}$ **-Wilson Current Source**
- $I_{C2} = I_O = \frac{I_{REF}}{(1+\frac{2}{\beta(1+\beta_3)})}$

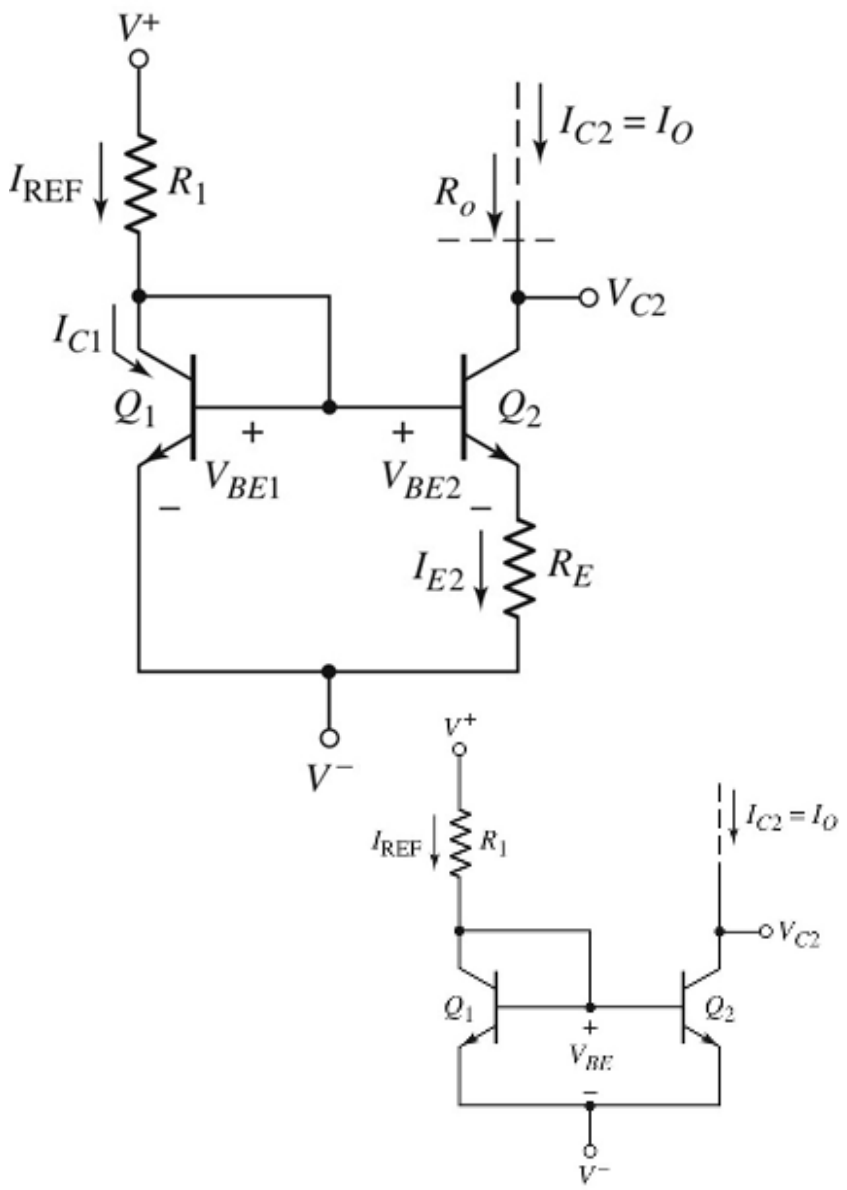
------ Three transistor current source

- $I_{C2} = \frac{I_{REF}}{(1+\frac{2}{\beta})}$ ------Two transistor current source

Difference in Wilson and Three Transistor Current Source

- Output Resistance
- $R_{O\ Wilson} = (\beta\ r_{O3})/2 > R_{OTwo\ transistor}$ or $R_{OThree\ transistor}$
- $\Rightarrow$ In Wilson current source , change in bias current Io with change in output collector voltage is much smaller

Widlar Current Source



In the current source circuit considered so far, the load and reference currents have been nearly equal.

For two transistor current source, if we require load current of I_O = 10 μA , then for V^+ =5 V, V^- = -5 V, required resistance value,

$R_1 = (V^+ – V_{BE} – V^-)/ I_{REF}$

$= 5\text{-}0.7\text{-}(-5)/ 10 = 930\ K\Omega$

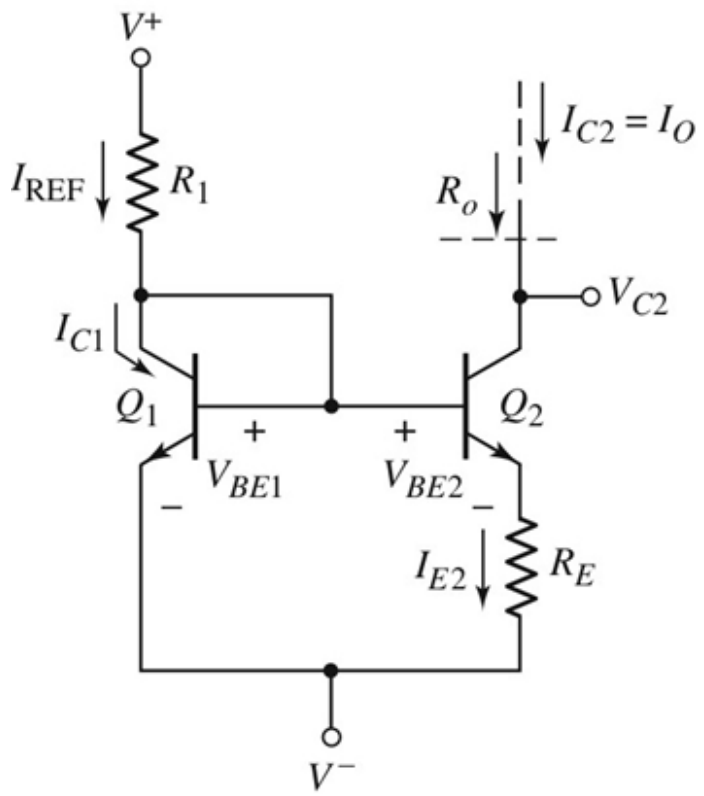
In IC , resistors on the order of 1 MΩ require larger area and are difficult to fabricate accurately. Need to limit IC resistor to low KΩ range.

Widlar Current Source

- Widlar Current Source meet this objective.
- Voltage difference is produced across R_E, so that B-E voltage of Q2 is less than B-E voltage of Q1. A smaller B-E Voltage produces a smaller collector current.
- $\Rightarrow$ Load current I_O is less than reference current I_{REF}.

Current relationship I_O & I_{REF}

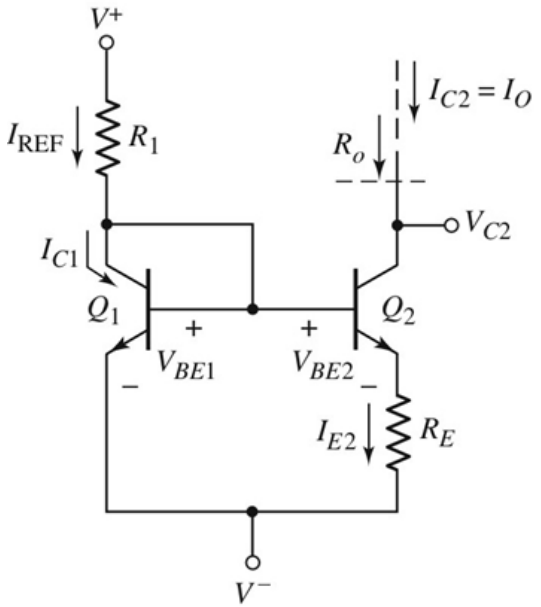
- Transistors are identical. $\beta_1 = \beta_2 = \beta \gg 1$



- $I_{REF} \cong I_{C1} = I_S e^{\frac{V_{BE1}}{V_T}}$
- $I_O = I_{C2} = I_S e^{\frac{V_{BE2}}{V_T}}$
- $V_{BE1} = V_T \ln (I_{REF}/ I_S)$
- $V_{BE2} = V_T \ln (I_O/ I_S)$
- $V_{BE1} - V_{BE2} = V_T \ln (I_{REF}/ I_O)$
- From Circuit,
- $V_{BE1} - V_{BE2} = I_{E2} R_E \cong I_O R_E$
- $I_O R_E = V_T \ln (I_{REF}/ I_O)$

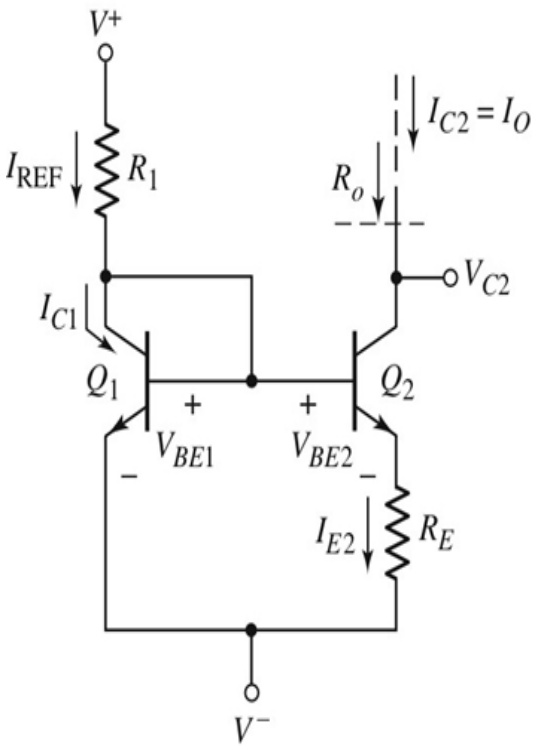
Design Widlar current source to achieve specified reference and load current

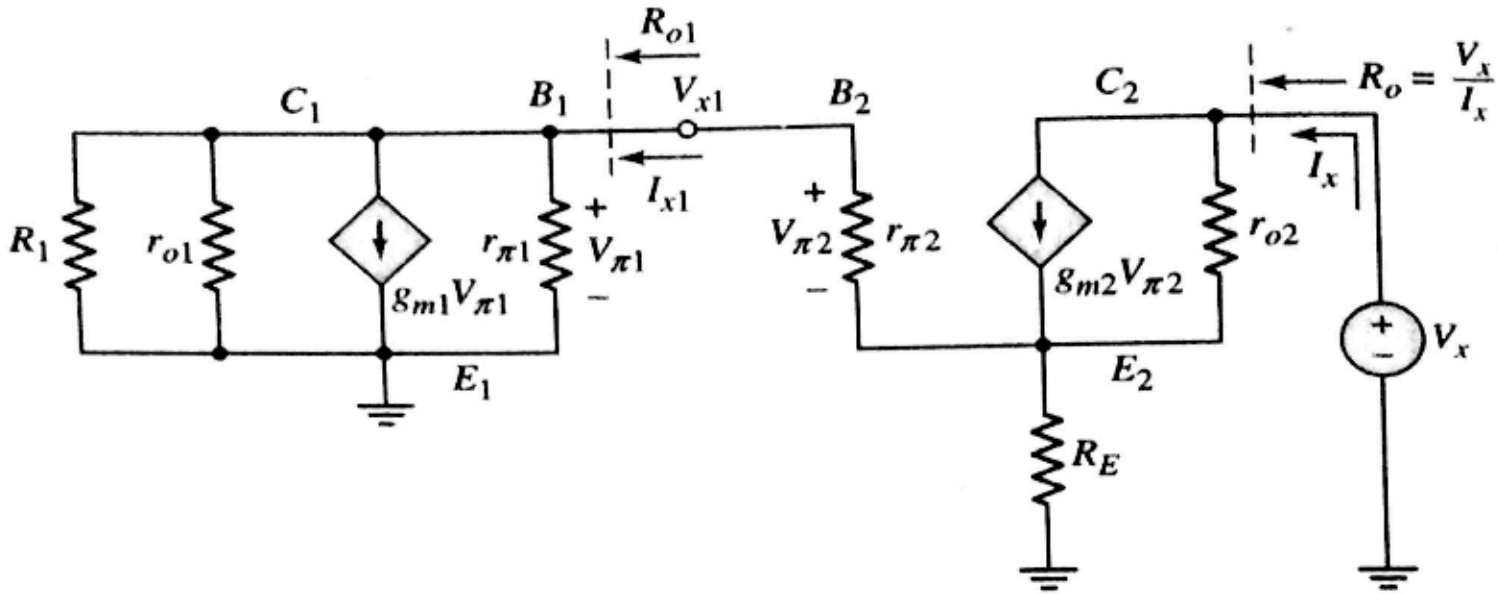
- I_O = 12 μA , I_{REF} = 1 mA, V^+ =5 V, V^- = -5 V, V_{BE1} = 0.7 V
- $R_1 = (V^+ - V_{BE} - V^-)/ I_{REF}$ =5- 0.7-(-5)/1 = 9.3 KΩ
- $R_E = (V_T / I_O) \ln (I_{REF}/ I_O)$ = 9.58 KΩ



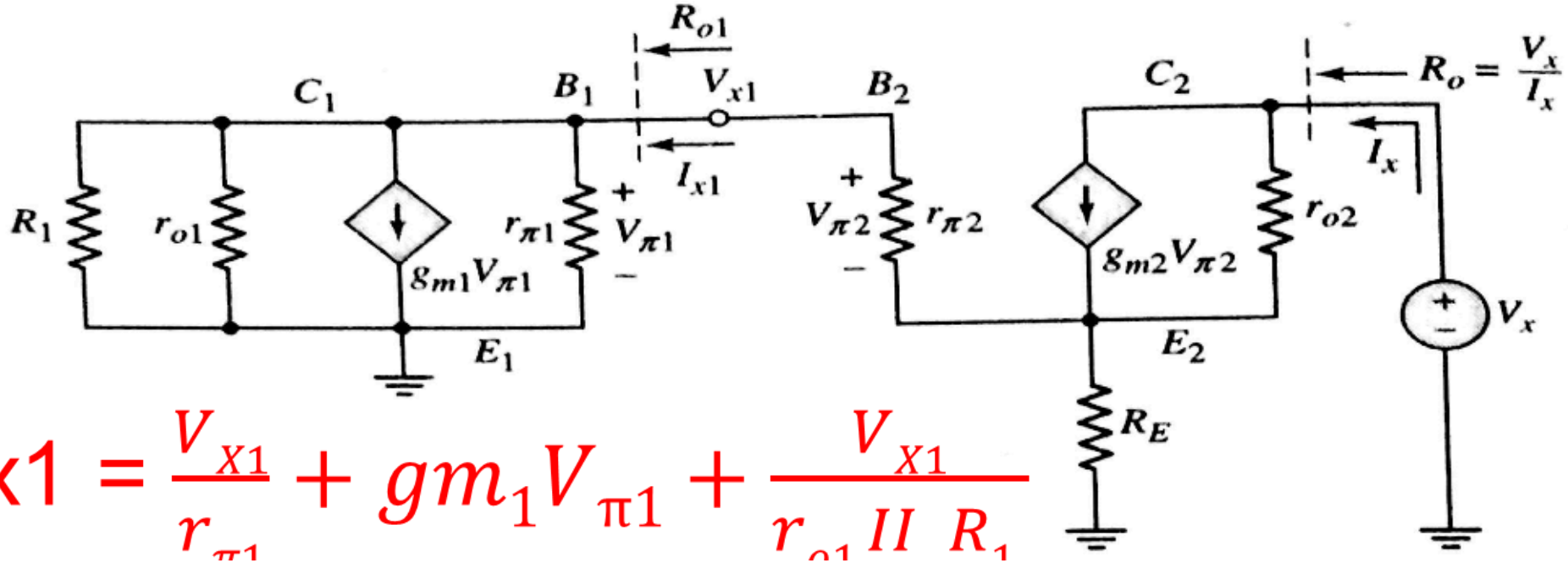
Output Resistance

- Change in load current with change in Voltage V_{C2} of Widlar current source can be expressed as $\frac{dI_o}{dV_{C2}} = \frac{1}{R_o}$ R_O is output resistance looking into collector of Q2.



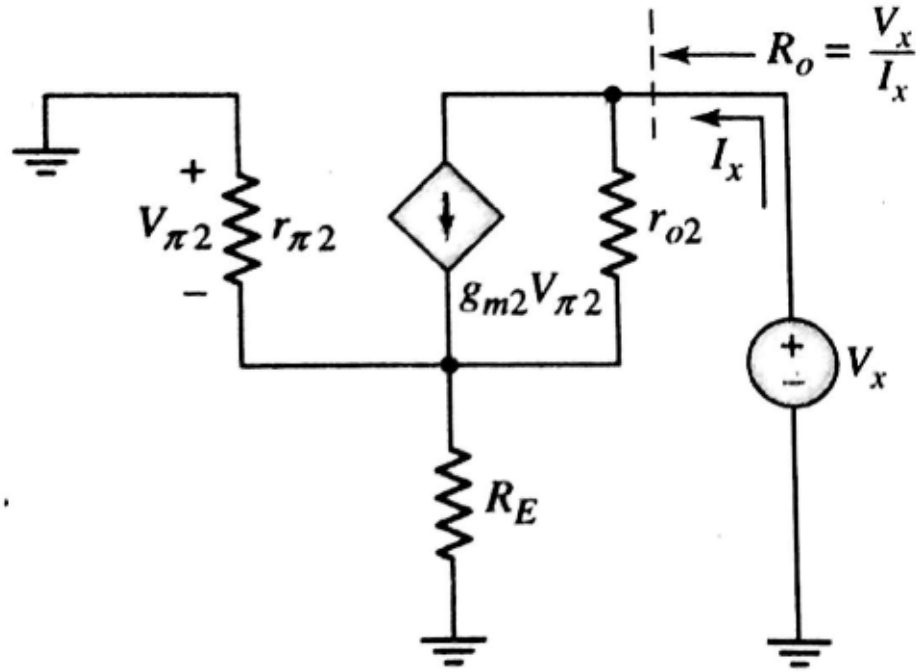


(a)

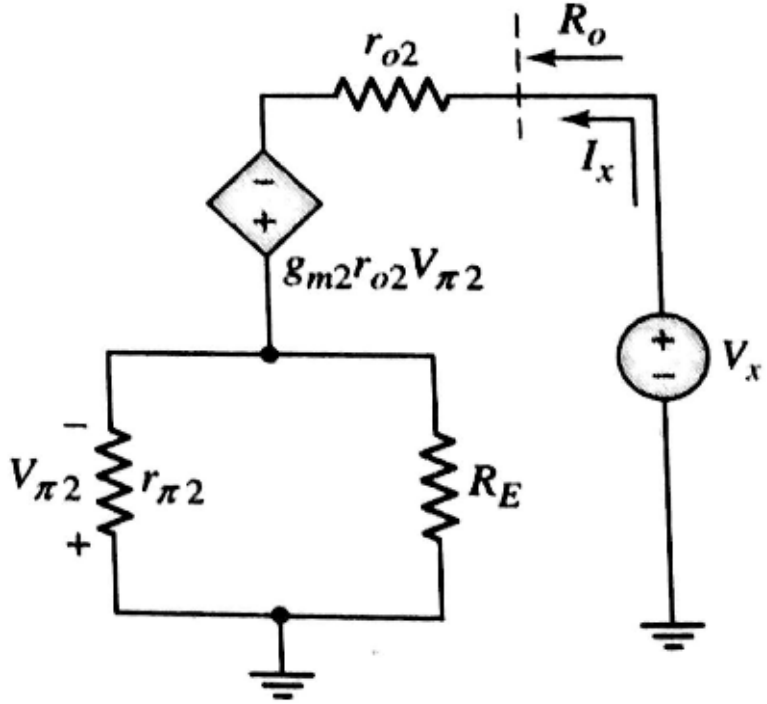


$$Ix1 = \frac{V_{X1}}{r_{\pi 1}} + gm_1 V_{\pi 1} + \frac{V_{X1}}{r_{o1}\ II\ R_1}$$

(a)

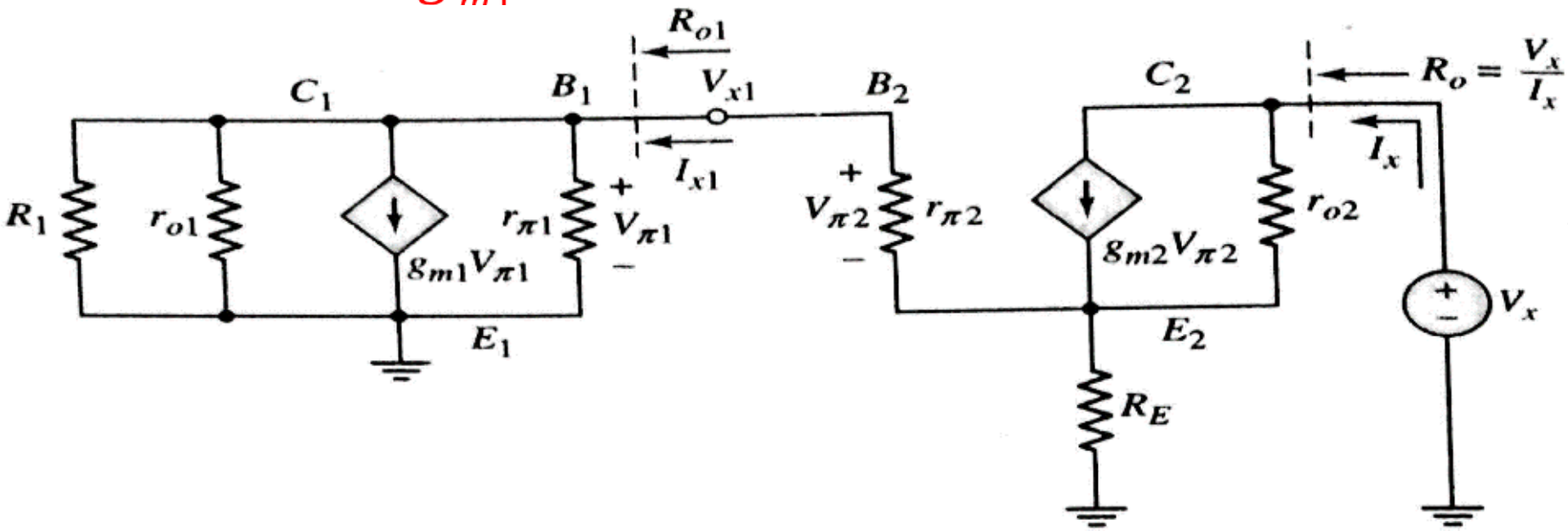


(b)



(c)

- $I_{X1} = \frac{V_{X1}}{r_{\pi 1}} + gm_1 V_{\pi 1} + \frac{V_{X1}}{r_{O1} \; II \; R_1}$
- $V_{\pi 1} = V_{X1}$
- $\frac{1}{R_{O1}} = \frac{I_{X1}}{V_{X1}} = \frac{1}{r_{\pi 1}} + gm_1 + \frac{1}{r_{O1} \; II \; R_1}$
- OR
- $R_{O1} = r_{\pi 1} II \frac{1}{g_{m1}} II r_{O1} II \; R_1$

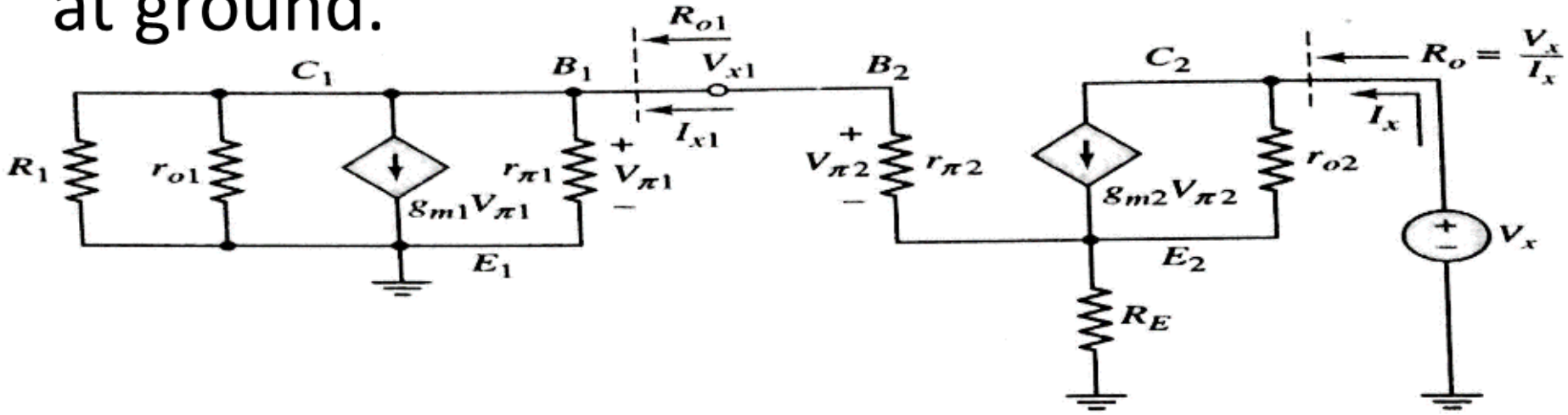


(a)

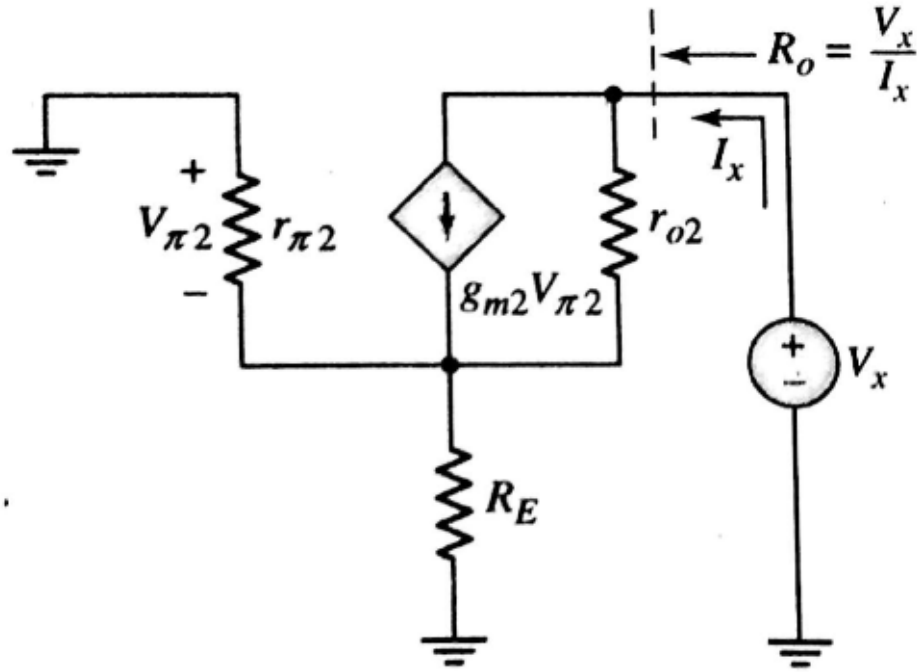
- V_{BE} =0.7V, β= 100, V^{+} =3V, V^{--} = -3V, I_{REF} =1mA, R_1 =9.3K, V_A = ∞ Io = 12 micro A
- $r_{\pi 1}$ = 2.6K g_{m1} = 38.5mA/V
- R_{O1} = 26 ohm

$$R_{O1} = r_{\pi 1} II \frac{1}{g_{m1}} II r_{O1} II \; R_1$$

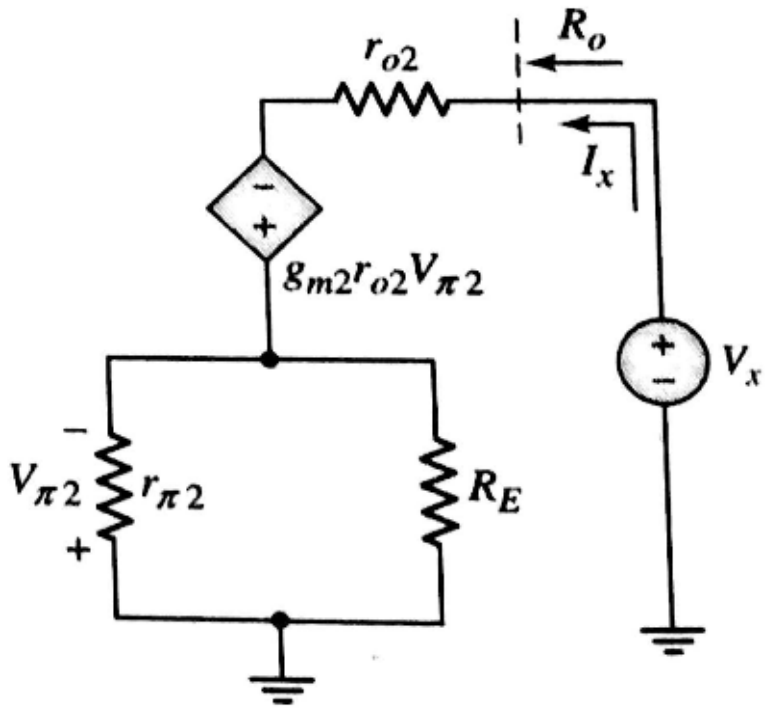
- Io = 12 μ A, so $r_{\pi 2}$ = 217 K
- R_{O1} *is in series with* $r_{\pi 2}$
- $R_{O1} \ll r_{\pi 2}$, So , neglect R_{O1} . So base of Q2 is at ground.



(a)



(b)



(c)

$V_{\pi 2} = - I_x (R_E // r_{\pi 2})$ ---1

$V_x = I_x r_{O2} - g_{m2} V_{\pi 2} r_{O2} + I_x R_E'$ ---2

From 1 & 2

$\frac{V_x}{I_x} = (r_{O2} + g_{m2} R_E' r_{O2} + R_E')$

Ro $= r_{O2} (1 + R_E' [g_{m2} + 1/r_{o2}])$

$1/r_{o2} \ll g_{m2}$ So, **Ro** $= r_{O2} (1 + R_E' g_{m2})$

Problem-Solving Technique: BJT Current Source Circuits

1. Sum currents at various nodes to find relation between reference and bias currents
2. Place test voltage at output node and analyze small-signal equivalent circuit to find output resistance.